

Reasoning Ctrl+F

Innovative Document Navigation: Reasoning-based Search

Group 1

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Existing Ctrl+F



- Does NOT understand
- Should know exact word
- No room for other language

Motivation

**What if... we can build a
Ctrl+F that can think?**

Data

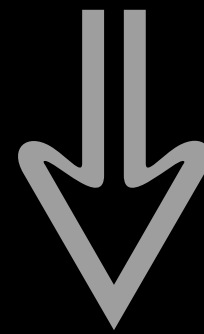


WIKIPEDIA

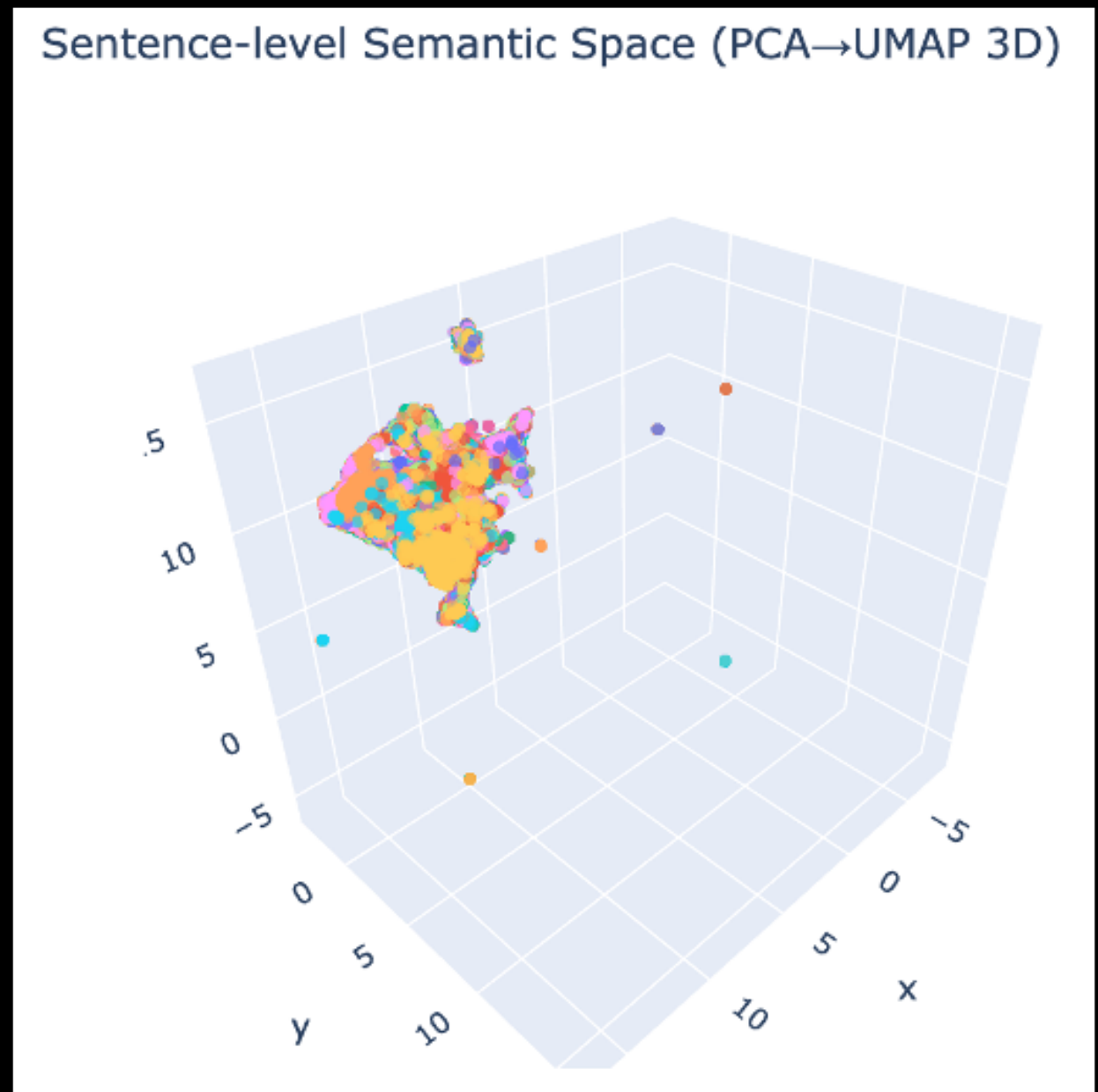
Wikipedia dataset – all countries / top 1000 cities

Data

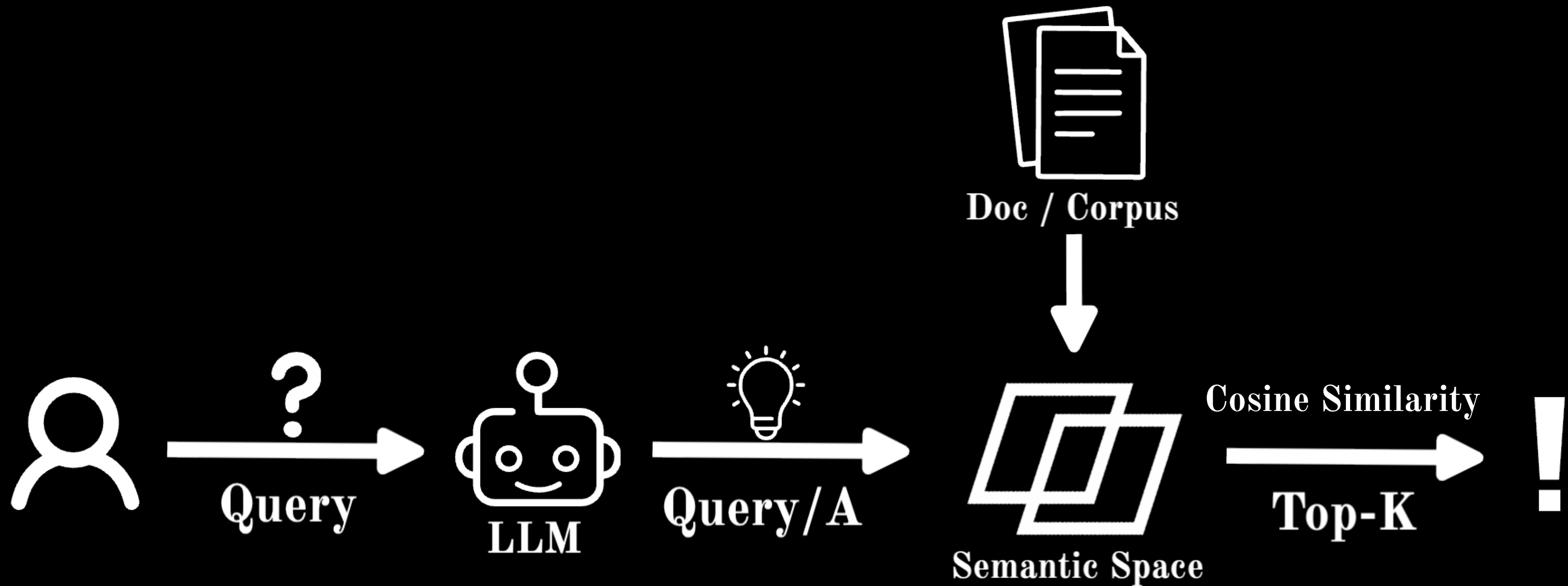
- In a official language
- Various types of sentences
- Highly structured
- Decent credibility
- Expressional Diversity
- No slang or emojis



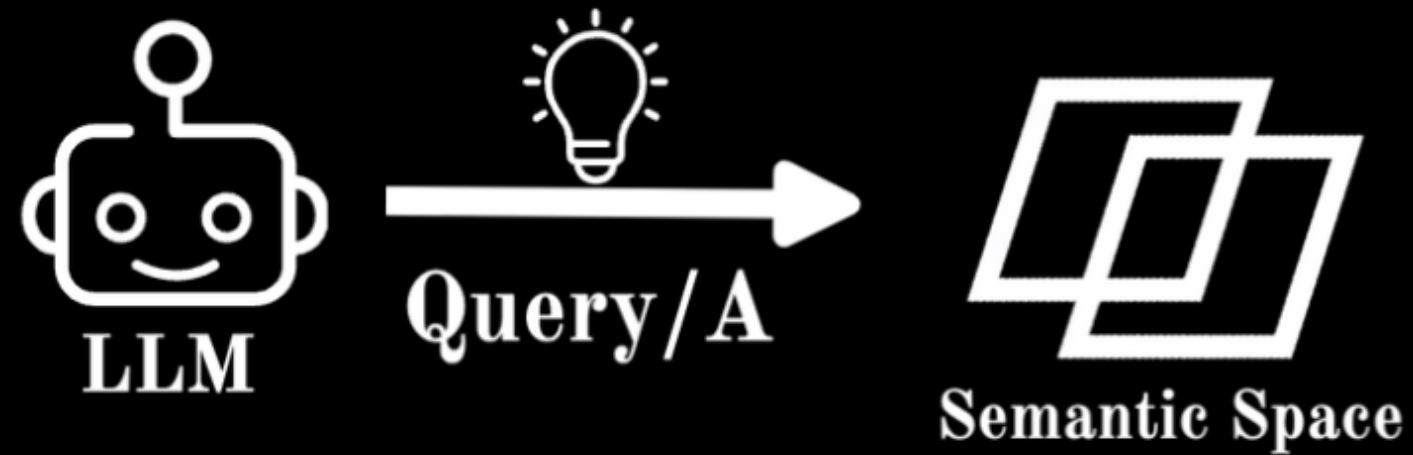
Standard preprocessing would suffice



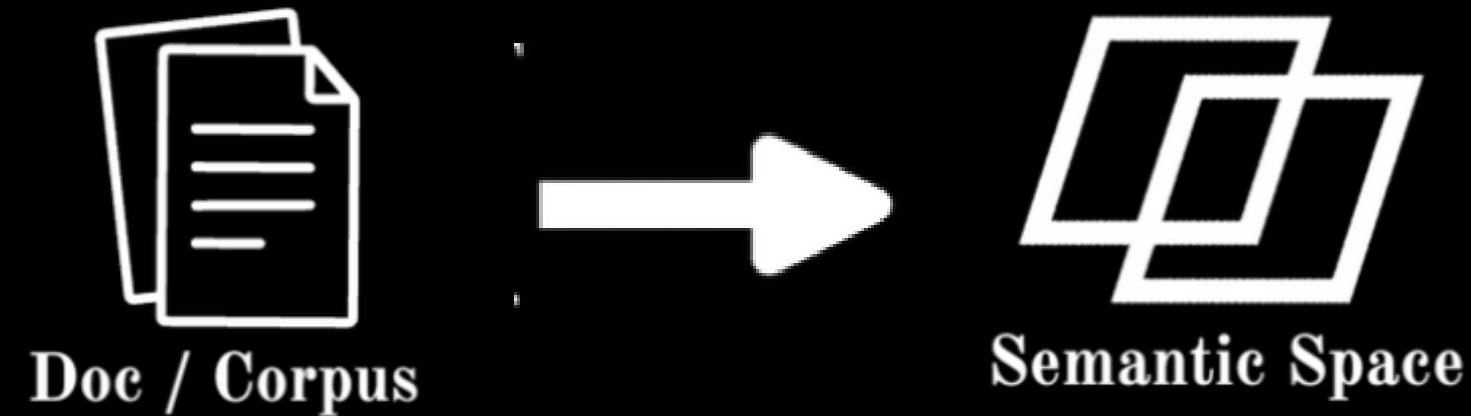
Methodology



Methodology



Which LLM? GPT
Feed Query+Answer



Which embedding?
SimCSE

Methodology

Why Top-k?

Our original intention aligns with the top-p approach along with thresholding...

**Still, cosine similarity does NOT
mean correctness**

+) Difficult Threshold Calibration

Methodology

Likert Scale – Qualitative Evaluation

Score	Description
1	All entities and relations correct
2	Entities correct, minor relational omission
3	Partial entity match, relation missing
4	Only superficial topical match
5	Completely irrelevant or contradictory

AI / Human
Evaluation

Ablation Studies

Our Best Combination so far..

- **SimCSE**

Cosine Similarity

0.7926

- **GPT**



Likert Scale

- **Query + Answer**

4 / 3.92

for 100 average results

Ablation Studies

Embedding method

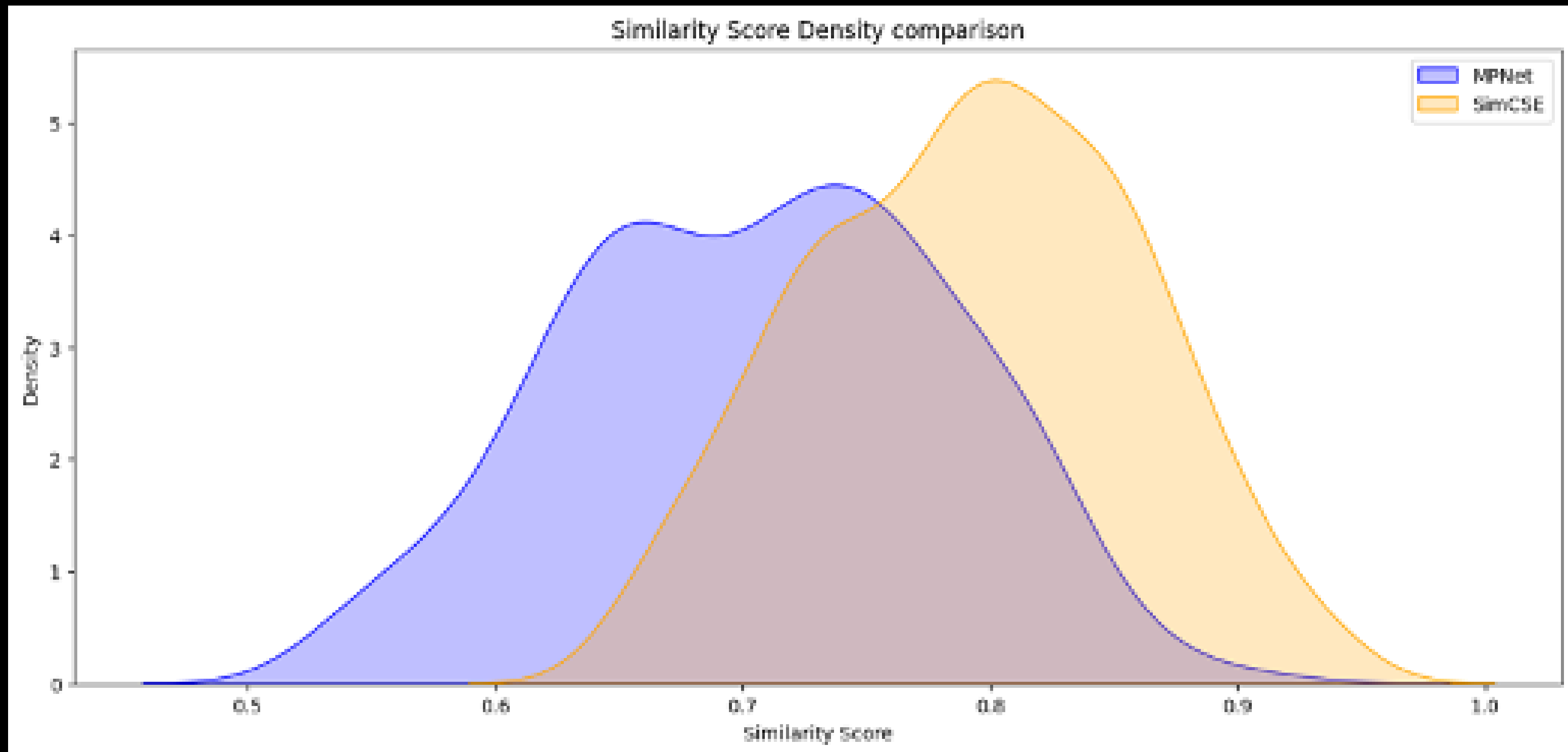
(Gemini)	TF-IDF	MPNet	SimCSE
Similarity	0.4555	0.7483	0.8112
Likert (AI)	.	3.12	3.78
Likert (Human)	.	2.68	3.52

TF-IDF vs Dense Methods

for 50 average results

Ablation Studies

- **SBERT vs SimCSE**



Ablation Studies

Input Query Type

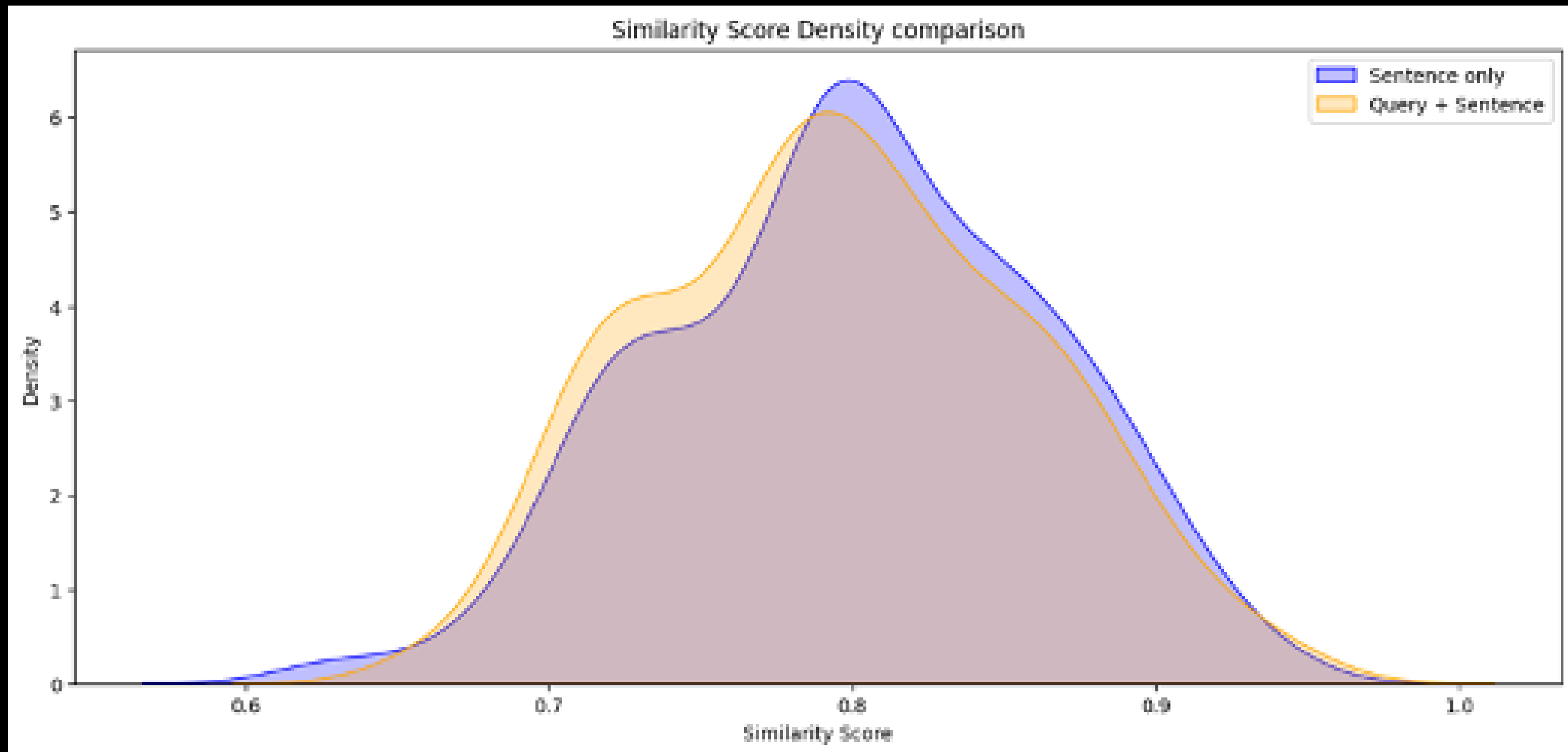
(SimCSE, Gemini)	LLM-only	Query+LLM
Similarity (100 avg)	0.7852	0.8023
Likert (AI)	3.4	3.92
Likert (Human)	3.56	3.87

LLM-Ans vs Query + LLM-Ans

for 100 average results

Ablation Studies

- **LLM-A vs Query + LLM-A**



Ablation Studies

LLM Type

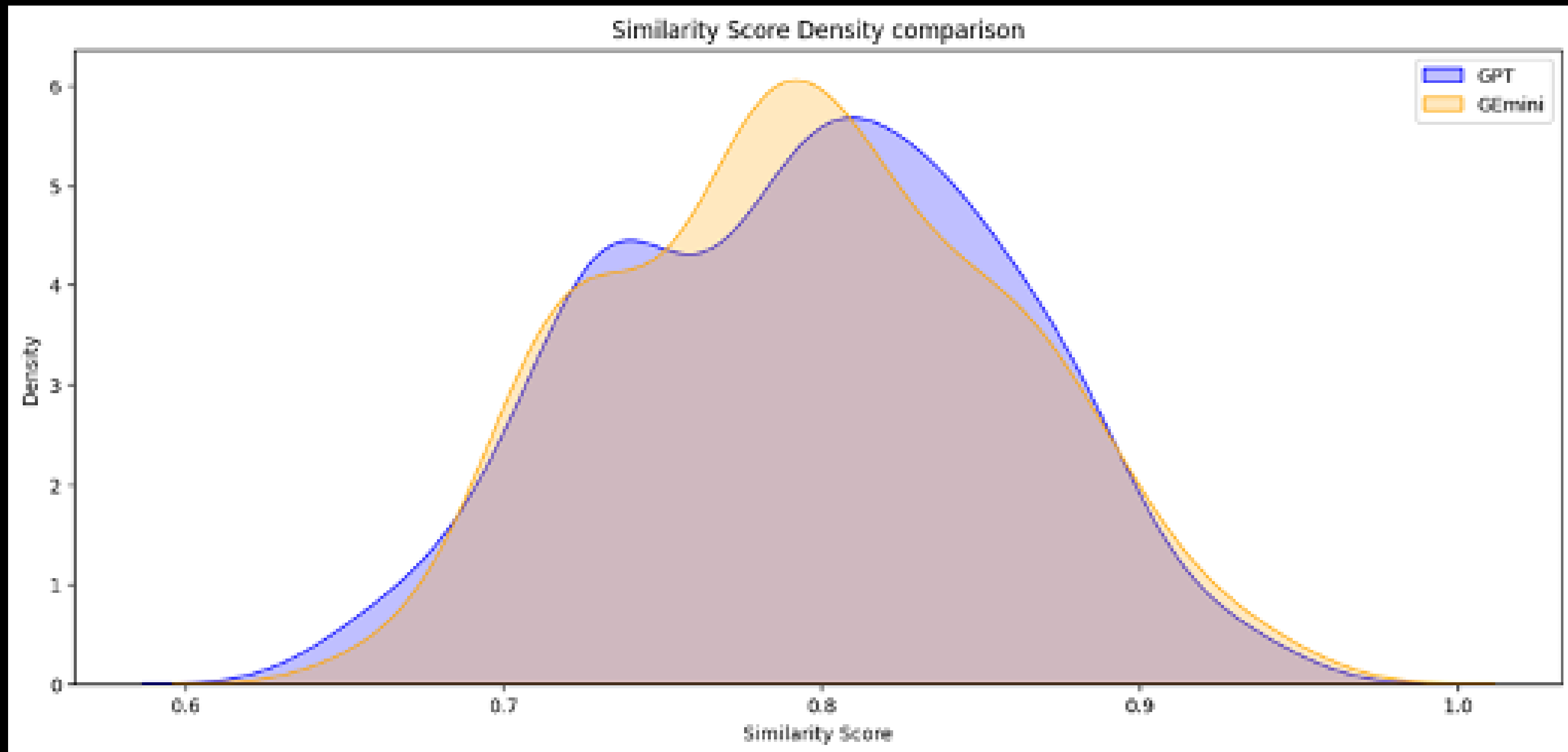
(Query + LLM, SimCSE)	Gemini	GPT
Similarity (100 avg)	0.8023	0.7926
Likert (AI)	3.36	4
Likert (Human)	3.28	3.92

Gemini vs GPT

for 50 average results

Ablation Studies

- Gemini vs GPT



Ablation Studies

Reflections

- **Query + LLM-Ans**

- Help catching relational elements
- Preserve lexical anchor (specificity + coherence)
- Help focusing on key part

- **SimCSE**

- Sentences centered latent-space
- vs SBERT: different focus
- Rather patternized structure and clear entities & relationships

Lessons

- **Importance of Evaluation**
- **Abundance of Unexpectedness**
- **No Dominance of Advanced Techniques**
- **Hard to set retrieval threshold**

It will never work as we intended

Thanks!

Questions or Feedbacks are more than welcome

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